Zeolites have porous structure in which the monovalent sodium ions and water molecules are loosely held. The Si and Al atoms are tetrahedrally coordinated with each other through shared oxygen atoms. Zeolites are similar to clay minerals but they differ in their crystalline structure.

Zeolites have a three dimensional crystalline structure looks like a honeycomb consisting of a network of interconnected tunnels and cages. Water molecules moves freely in and out of these pores but the zeolite framework remains rigid. Another special aspect of this structure is that the pore/channel sizes are nearly uniform, allowing the crystal to act as a molecular sieve. We have already discussed in XI standard, the removal of permanent hardness of water using zeolites.



Boron Neutron Capture Therapy:

The affinity of Boron-10 for neutrons is the basis of a technique known as boron neutron capture therapy (BNCT) for treating patients suffering from brain tumours.

It is based on the nuclear reaction that occurs when boron-10 is irradiated with low-energy thermal neutrons to give high linear energy α particles and a Li particle.

Boron compounds are injected into a patient with a brain tumour and the compounds collect preferentially in the tumour. The tumour area is then irradiated with thermal neutrons and results in the release of an alpha particle that damages the tissue in the tumour each time a boron-10 nucleus captures a neutron. In this way damage can be limited preferentially to the tumour, leaving the normal brain tissue less affected. BNCT has also been studied as a treatment for several other tumours of the head and neck, the breast, the prostate, the bladder, andthe liver.

Summary

- The elements in which their last electron enters the 'p' orbital, constitute the p-block elements.
- The p-block elements have a general electronic configuration of ns^2, np^{1-6}. The elements of each group have similar outer shell electronic configuration and differ only in the value of n (principal quantum number).
- Generally on descending a group the ionisation energy decreases and hence the metallic character increases.
- The ionisation enthalpy of elements in successive groups is higher than the corresponding elements of the previous group as expected.
- As we move down the 13th group, the electronegativity first decreases from boron to aluminium and then marginally increases.

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